Question # 1 :-

1. if $\vec{A}$ and $\vec{B}$ are perpendicular, then $\vec{A} \cdot \vec{B} = 0$, so,

$(6 \times 4) + (-1 \times x) + (\sqrt{9} \times -2) = 0$

$24 - x - 6 = 0$

$+x = +18$

$\boxed{x = 18}$

Vector $\vec{b}$ is perpendicular to $\vec{A}$ if $x = 18$.

2. Dot product

- The elements corresponding to same row and column are multiplied together and the products are added.
- Result: scalar
- Order of vectors does not matter, just the number of elements in both vectors should be equal.

Hadamard product

- The elements corresponding to same row and columns of given vectors are multiplied together.
- Result: vector
- The order of vectors should be ~~the~~ same and the resulting vector will be of same order.

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3. Prove using vector projection & L-norm:

=> We know that dot product is;

$= \vec{u} \cdot \vec{v} = \|\vec{u}\| \cdot \|\vec{v}\| \cos(\theta)$ ___ (i)

=> Projection of $\vec{u}$ onto $\vec{v}$ is defined as;

$= \|\vec{u}_{proj}\| = \|\vec{u}\| \cos(\theta)$ ___ (ii)

=> Substituting (ii) in (i);

$= \vec{u} \cdot \vec{v} = \|\vec{u}_{proj}\| \cdot \|\vec{v}\|$ ___ (iii)

=> We know that;

$= \vec{u} = (1, 2)$

$= \vec{v} = (2, 2)$

$= \vec{u}_{proj} = (1.5, 1.5)$

=> Substituting the values in (iii);

$= \vec{u} \cdot \vec{v} = \sqrt{1.5^2 + 1.5^2} \times \sqrt{2^2 + 2^2}$

$= \boxed{\vec{u} \cdot \vec{v} = 6}$

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